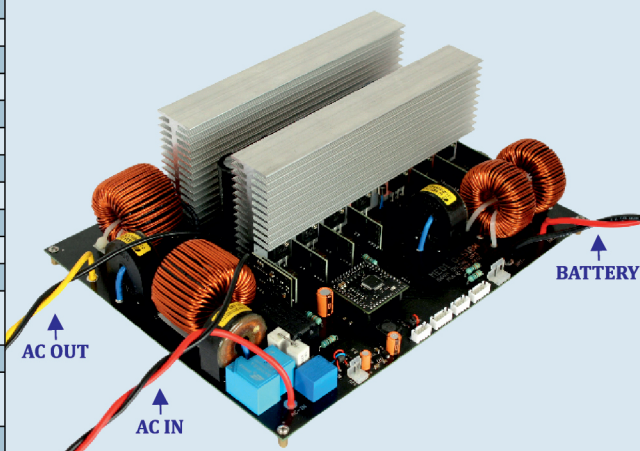


# MEDI's 5KVA ONLINE UPS Technology Transfer

## MLS (Multi Level Switching) TECHNOLOGY with APFC

High frequency Transformer-less UPS with front end PFC, harmonic reduction, common neutral, high efficiency three level inverter stage

|                                          |                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Capacity                                 | 1KVA, 2KVA, 3KVA, 4KVA, 5KVA<br>single phase input, single phase output                                                                  |
| Nominal AC input                         | 230V / 110V                                                                                                                              |
| AC input range                           | 170V to 270V / 80V to 130V                                                                                                               |
| Input frequency                          | 50Hz / 60Hz                                                                                                                              |
| Input power factor                       | >0.98 (full load)                                                                                                                        |
| Input current harmonics                  | < 4% (full load)                                                                                                                         |
| Current crest factor ratio               | 3:1 max                                                                                                                                  |
| Nominal output voltage                   | 230V / 110V                                                                                                                              |
| Output frequency                         | 50Hz / 60Hz                                                                                                                              |
| Output voltage regulation                | < +/-1%                                                                                                                                  |
| Output waveform                          | Pure sinewave                                                                                                                            |
| Output voltage harmonic distortion (THD) | < 2%                                                                                                                                     |
| Battery voltage                          | 96V to 300V single battery bank                                                                                                          |
| Transfer time                            | Mains to battery - 0ms<br>Battery to mains - 0ms                                                                                         |
| Automatic bypass                         | Static bypass using SCR                                                                                                                  |
| Topology                                 | True online with double conversion with high efficiency three level inverter and APFC. Neutral common, no isolation transformer required |
| Battery type                             | Lithium / lead acid batteries                                                                                                            |
| Charging current                         | 1 to 15A settable                                                                                                                        |
| Protection                               | Overload, surge load, short circuit, battery low, mains low voltage, mains high voltage, high temperature                                |



**BOM of 3KVA ONLINE UPS**  
**Rs. 9800/-**

**Technical know-how cost**  
**upto 5KVA Rs. 7,50,000/-**

### Features –

- Input front end PFC to correct the current waveform to pure sine wave & power factor will be close to unity 0.98
- Three level inverter stage results in very low switching loss and requires compact magnetics.
- Inverter stage and the PFC stage will process only the difference power which makes the efficiency very high and light weight.
- Absolute transformer-less design - no iron core transformer and no ferrite transformer used. Only light weight high frequency inductors are used. This results in very easy manufacturing.
- Suitable for discrete MOSFET & IGBT
- Static bypass for uninterruptable bypass during hazard condition.
- 4-level battery charging.
- Charging current up to 15A Suitable for Lithium batteries.
- 4-line LCD and graphic display.
- Wifi connectivity with Android.
- Temperature sensor and fan control.
- Suitable for frequency converters.

For live demo, visit us at booth # H11. L 45 at Electronica India Expo 2024, Greater Noida, Delhi NCR on 11th - 13th September 2024

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